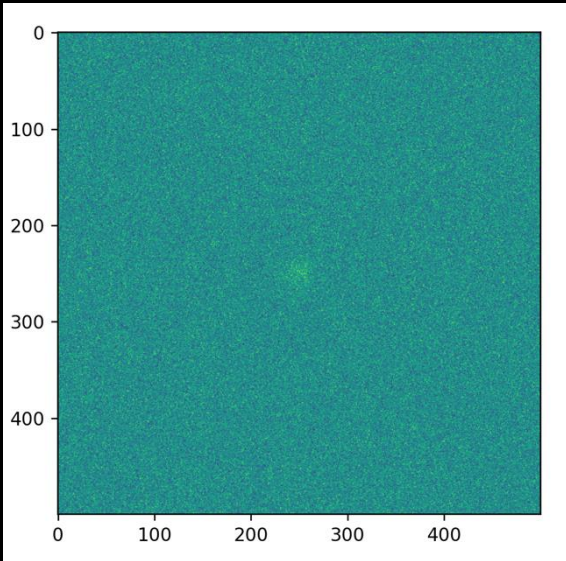
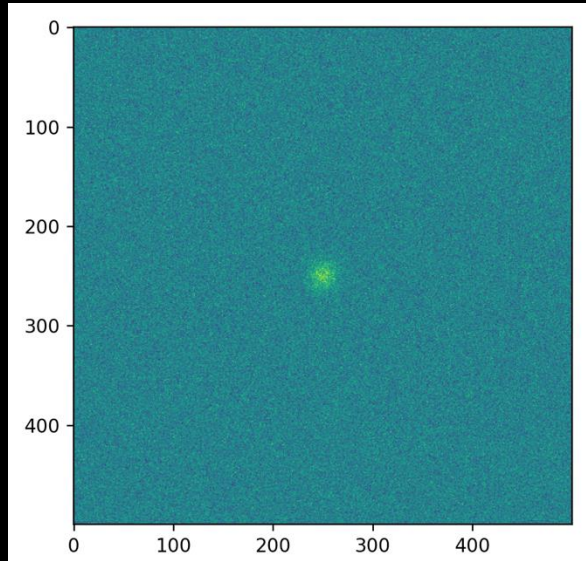


ASTR 302 Fall 2026

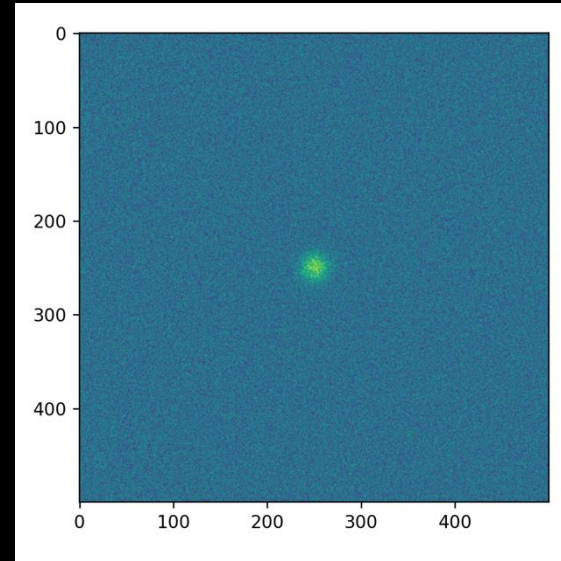
Estimating SNR and Time Required for Observing



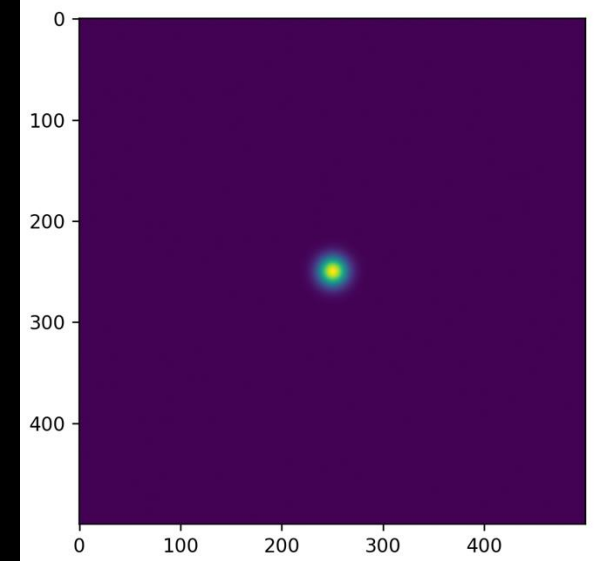
$F = 10$
 $\sigma = 10$
SNR ~ 1



$F = 30$
 $\sigma = 10$
SNR ~ 3



$F = 50$
 $\sigma = 10$
SNR ~ 5



$F = 10000$
 $\sigma = 10$
SNR ~ 1000

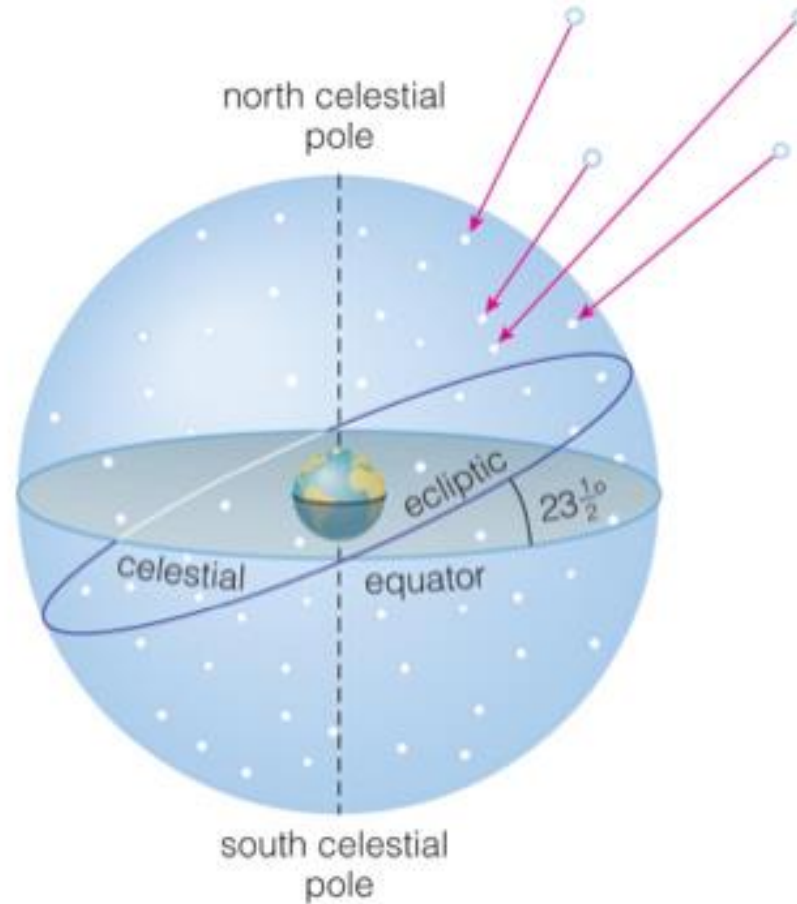
Agenda – September 9, 2026

- Locating Objects in 3-D Space
- Error and the Signal to Noise Equation
- Estimating required exposure time in photon noise limit
- Scaling required time between telescopes

Reference Material:

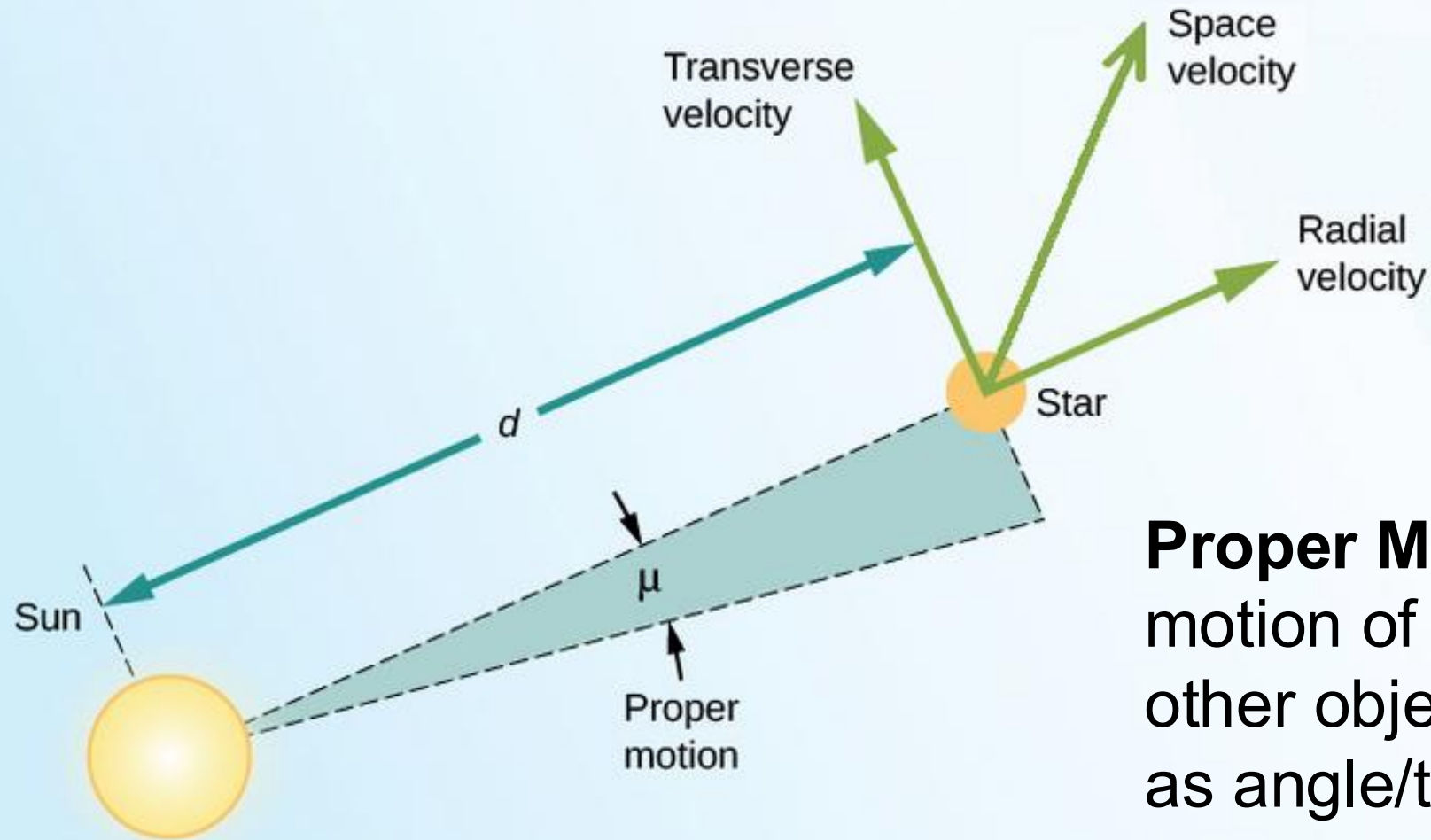
Applying Taylor Chapter 3, Birney Chapter 9

Where Things Are **Located** is Just as Important as What They Are



Astrometry is the science of measuring the positions and motions of objects on the celestial sphere.

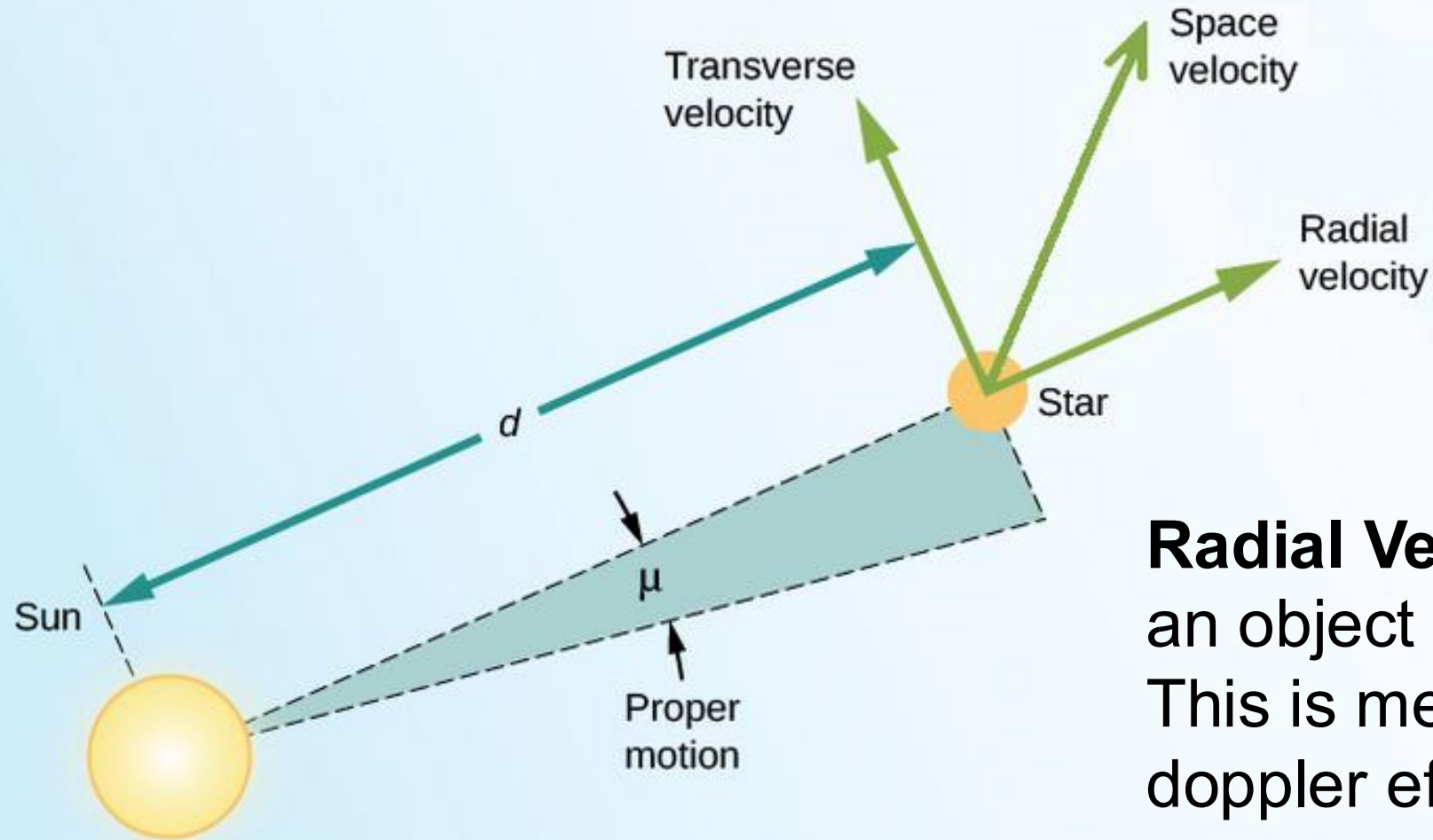
Proper Motion



Proper Motion is the angular motion of a star/target relative to other objects. Usually reported as angle/time.

Proper Motion and Velocity of a Star

Radial Velocity



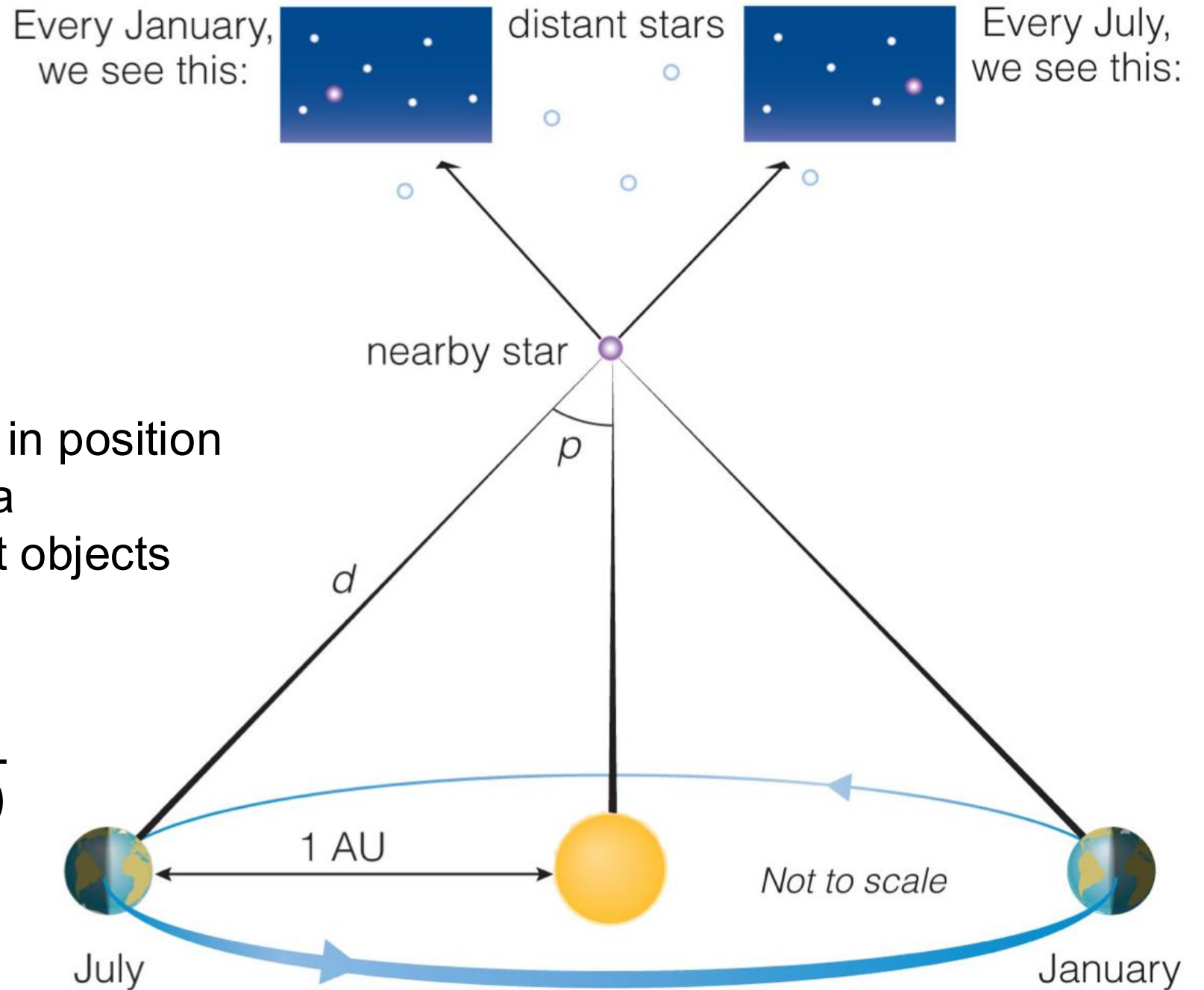
Radial Velocity is the motion of an object along our line of sight. This is measured with the doppler effect (spectroscopy).

Proper Motion and Velocity of a Star

Parallax

Parallax the apparent shift in position of a nearby object against a background of more distant objects

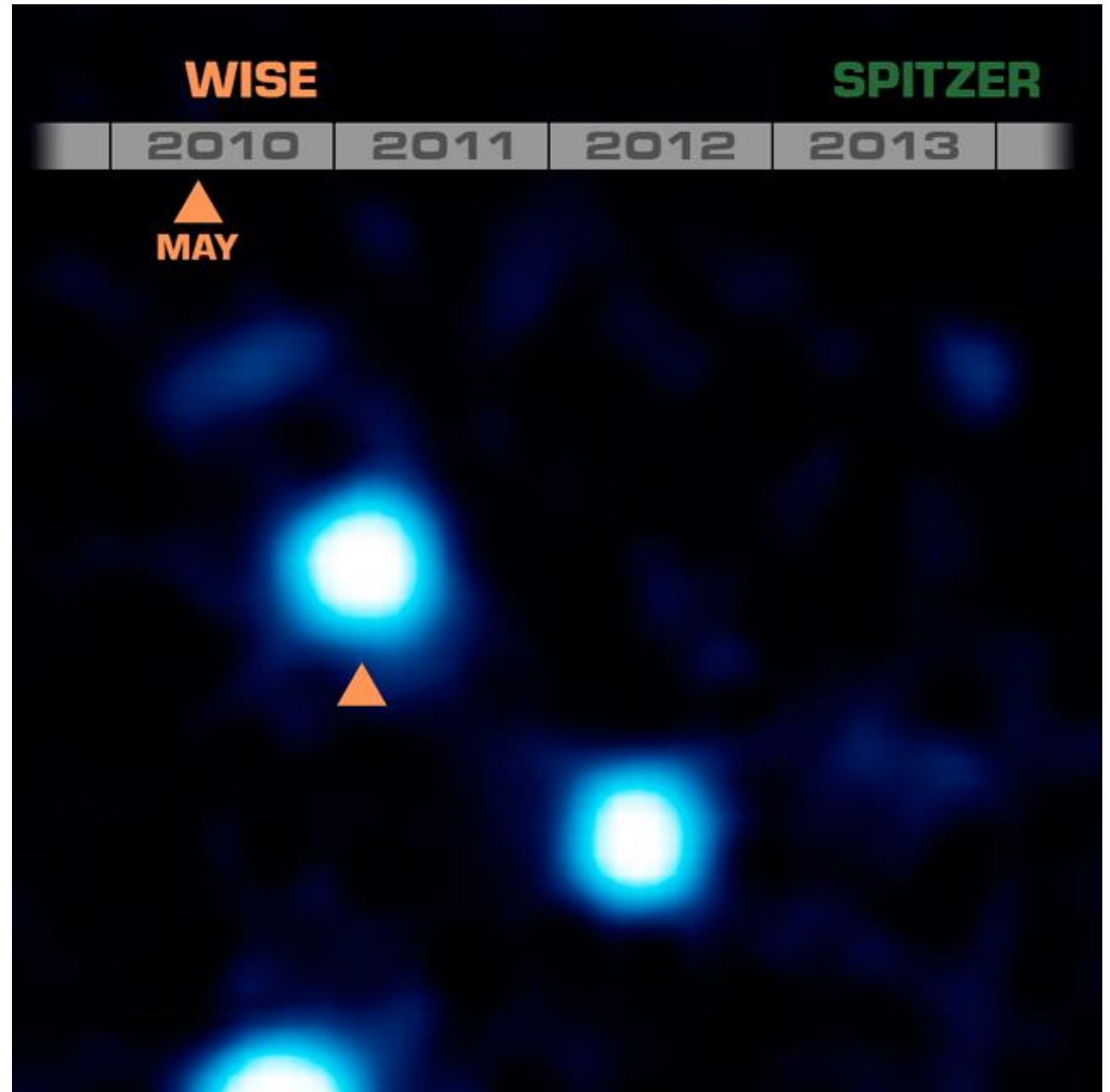
$$d(\text{pc}) = \frac{1}{\pi(\text{arcseconds})}$$



Parallax

Parallax the apparent shift in position of a nearby object against a background of more distant objects

$$d(\text{pc}) = \frac{1}{\pi(\text{arcseconds})}$$



WISE 0855 Animation of Figure 1. in Luhman+ 2014

Where Things Are **Located** is Just as Important as What They Are

To Locate Something in 3-D:

- **Right Ascension** and **Declination** (where something was located relative to the vernal equinox a given year)
- **Proper Motion** - How much your target moves each year relative to you.
- **Parallax** – The apparent shift of your target relative to background objects.
- **Radial Velocity** – an object's motion along your line of sight.

How Do Astronomers Estimate Errors

- **Error propagation** combines the individual errors from each variable that contributes to a derived quantity. This requires understanding where errors are introduced in your telescope and instrument.
- **Statistics** rely upon performing the experiment numerous times, where the value for the derived variable is the mean of the individual estimated values and the uncertainty is the standard deviation about that mean.

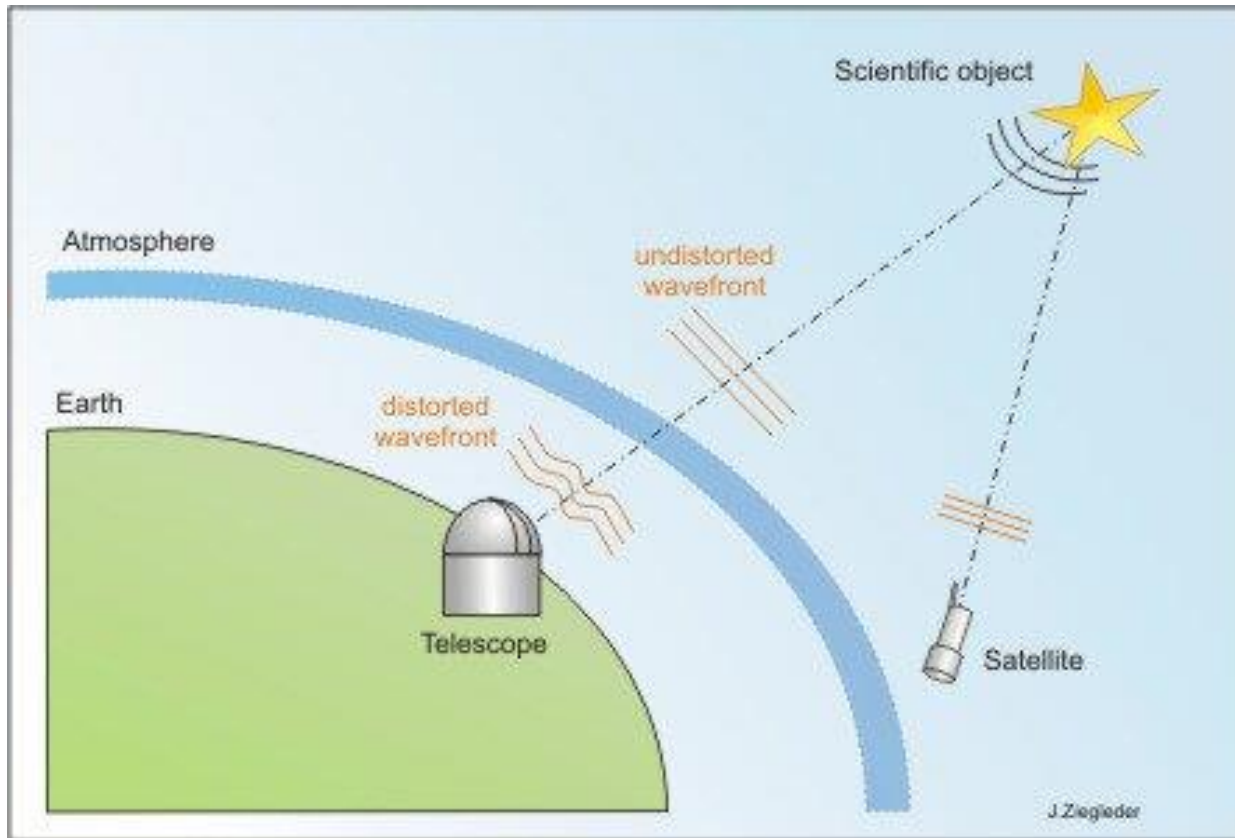
Error Propagation Formula

Uncertainty in a Function of Several Variables

Suppose that $x, \dots, z$ are measured with uncertainties $\delta x, \dots, \delta z$ and the measured values are used to compute the function $q(x, \dots, z)$. If the uncertainties in $x, \dots, z$ are independent and random, then the uncertainty in q is

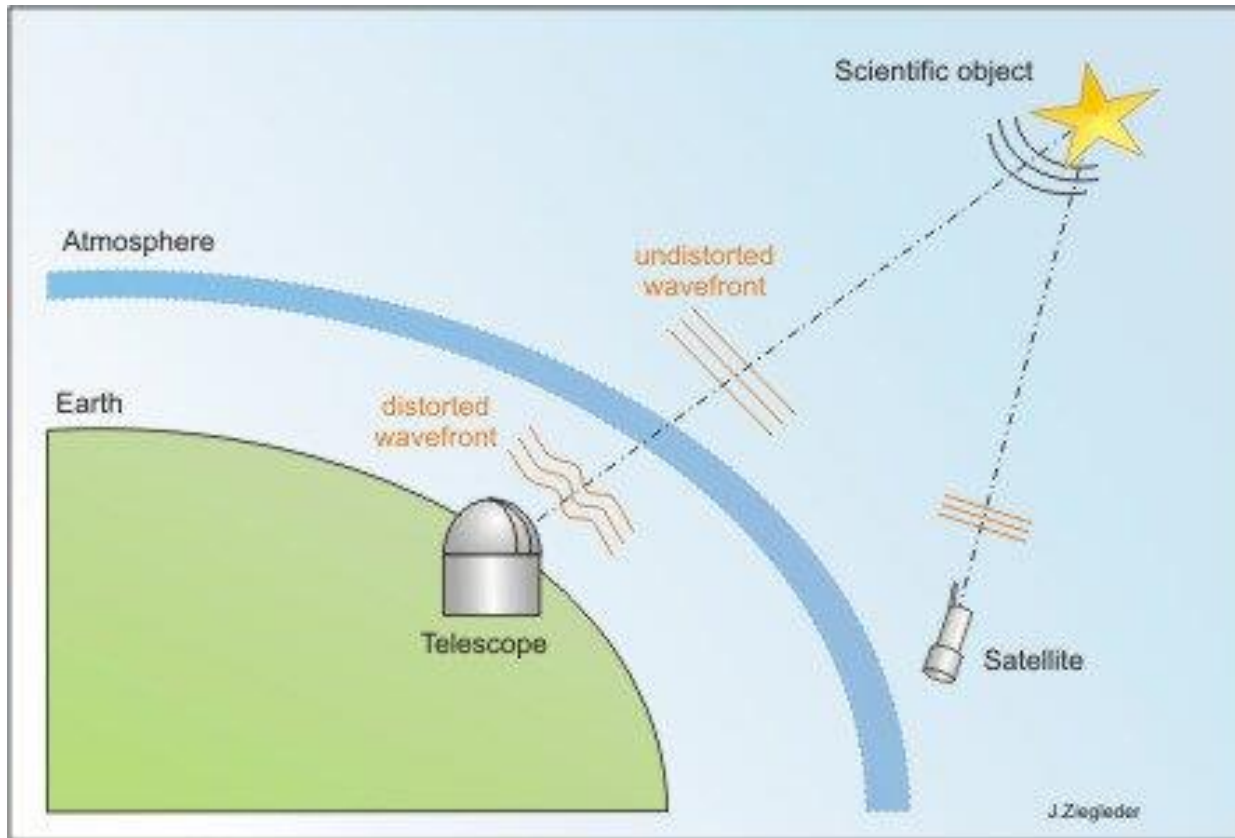
$$\delta q = \sqrt{\left(\frac{\partial q}{\partial x} \delta x\right)^2 + \dots + \left(\frac{\partial q}{\partial z} \delta z\right)^2}.$$

Sources of Error in Astronomical Observations



1. Random Noise from the Source
2. Earth's Atmosphere (Ground-based)
 - Sky/Airmass
 - Water Vapor (Infrared)
3. Optics/Materials from the telescope
4. Noise from your sensor/receiver/detector

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$$\text{Counts}_{\text{total}} = \text{Counts}_{\text{star}} + \text{Counts}_{\text{sky}} + \text{Counts}_{\text{Telescope}} + \text{Counts}_{\text{Detector}}$$

Estimating the Total Error

$$\text{Counts}_{\text{total}} = \text{Counts}_{\text{star}} + \text{Counts}_{\text{sky}} + \text{Counts}_{\text{Telescope}} + \text{Counts}_{\text{Detector}}$$

If we apply the standard error propagation formula, the total error is...

$$\sigma^2_{\text{total}} = \sigma^2_{\text{star}} + \sigma^2_{\text{sky}} + \sigma^2_{\text{telescope}} + \sigma^2_{\text{detector}}$$

$$\sigma_{\text{total}} = \sqrt{\sigma^2_{\text{star}} + \sigma^2_{\text{sky}} + \sigma^2_{\text{telescope}} + \sigma^2_{\text{detector}}}$$

Signal to Noise (SNR)

$$\text{SNR} = \frac{\text{Counts}_{\text{star}}}{\sigma_{\text{total}}}$$

Signal to Noise (SNR) Equation

$$SNR = \frac{gS_{star}}{\sqrt{gS_{star} + ngS_{sky} + ngS_{dark} + nS_{read}^2}}$$

S_{star} : Signal from the star (ADU)

S_{sky} : Signal from the sky (ADU)

S_{dark} : Signal from the dark (ADU)

S_{read} : Read noise (e-)

g : gain (e-/ADU)

n : Number of pixels

This equation does not include every possible error term, but the main ones for CCDs

Signal to Noise (SNR) Equation

$$SNR = \frac{gS_{star}}{\sqrt{gS_{star} + ngS_{sky} + ngS_{dark} + nS_{read}^2}}$$

Counts from source/star
Related to Flux/magnitude



S_{star} : Signal from the star (ADU)

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Counts from the sky

S_{star} : Signal from the star (ADU)

S_{sky} : Signal from the sky (ADU)

S_{dark} : Signal from the dark (ADU)

S_{read} : Read noise (e-)

g : gain (e-/ADU)

n : Number of pixels

Signal to Noise (SNR) Equation

$$SNR = \frac{gS_{star}}{\sqrt{gS_{star} + ngS_{sky} + ngS_{dark} + nS_{read}^2}}$$



S_{star} : Signal from the star (ADU)

S_{sky} : Signal from the sky (ADU)

S_{dark} : Signal from the dark (ADU)

S_{read} : Read noise (e-)

g : gain (e-/ADU)

n : Number of pixels

Read Noise from Detector

Signal to Noise (SNR) Equation

$$SNR = \frac{gS_{star}}{\sqrt{gS_{star} + ngS_{sky} + ngS_{dark} + nS_{read}^2}}$$

S_{star} : Signal from the star (ADU)

S_{sky} : Signal from the sky (ADU)

S_{dark} : Signal from the dark (ADU)

S_{read} : Read noise (e-)

g : gain (e-/ADU)

n : Number of pixels



Dark Current Noise
from Detector

Signal to Noise (SNR) in Photon Noise Limit

$$SNR = \frac{gS_{star}}{\sqrt{gS_{star} + ngS_{sky} + ngS_{dark} + nS_{read}^2}}$$

S_{star} : Signal from the star (ADU)

S_{sky} : Signal from the sky (ADU)

S_{dark} : Signal from the dark (ADU)

S_{read} : Read noise (e-)

g : gain (e-/ADU)

n : Number of pixels

For most of your projects, the targets are relatively bright

Signal to Noise (SNR) in Photon Noise Limit

$$SNR = \frac{gS_{star}}{\sqrt{gS_{star}}}$$


S_{star} : Signal from the star (ADU)

S_{sky} : Signal from the sky (ADU)

S_{dark} : Signal from the dark (ADU)

S_{read} : Read noise (e-)

g : gain (e-/ADU)

n : Number of pixels

For most of your projects, the targets are relatively bright

Signal to Noise (SNR) in Photon Noise Limit

$$\text{SNR} = \frac{\text{Counts}_{\text{star}}}{\sigma_{\text{total}}}$$

$$\text{SNR} = \sqrt{\text{Count Rate}_{\text{star}} \times t_{\text{exp}}}$$

For most of your projects, the targets are relatively bright

Signal to Noise (SNR) in Photon Noise Limit

$$\text{SNR} = \frac{\text{Counts}_{\text{star}}}{\sigma_{\text{total}}}$$

$$\text{SNR} = \sqrt{\text{Count Rate}_{\text{star}} \times t_{\text{exp}}} \sim \sqrt{\text{Flux}_{\text{star}} \times t_{\text{exp}}}$$

For most of your projects, the targets are relatively bright

Signal to Noise (SNR) in Photon Noise Limit

$$\text{SNR} = \sqrt{\text{Count Rate}_{\text{star}} \times t_{\text{exp}}} \sim \sqrt{\text{Flux}_{\text{star}} \times t_{\text{exp}}}$$

- By looking at brighter targets, you increase your SNR
- Exposing longer increases your signal to noise
- You need to observe 4 times as long just to double the signal to noise.
- Observing something half as bright reduces your signal to noise by 1/4

Flux and Magnitude

$$m - m_{ref} = -2.5 \log_{10} \left(\frac{F}{F_{ref}} \right).$$

- **Magnitude (m)** tells you how bright something is. The smaller the magnitude, the brighter something is. **Unitless.**
- **Flux (F)** also tells you how bright something is but in physical units. Higher values of flux means something is brighter. **Always has units.**

Example: Scaling Exposure Time

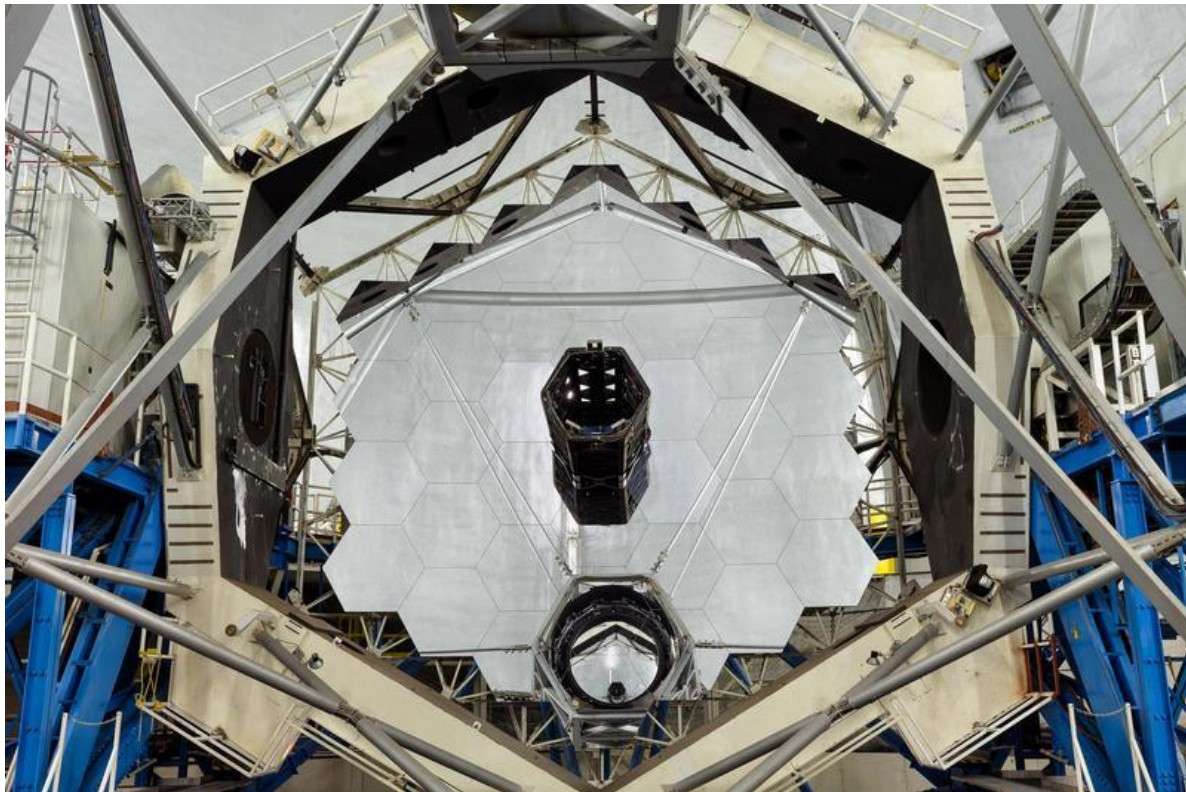
An instrument scientist tells you that they observed something V mag = 11 with a SNR of 1000 in 30 seconds. How long do you need to expose to achieve the same signal to noise on a V mag = 14 target?

$$\text{SNR} = \sqrt{\text{Count Rate}_{\text{star}} \times t_{\text{exp}}} \sim \sqrt{\text{Flux}_{\text{star}} \times t_{\text{exp}}}$$

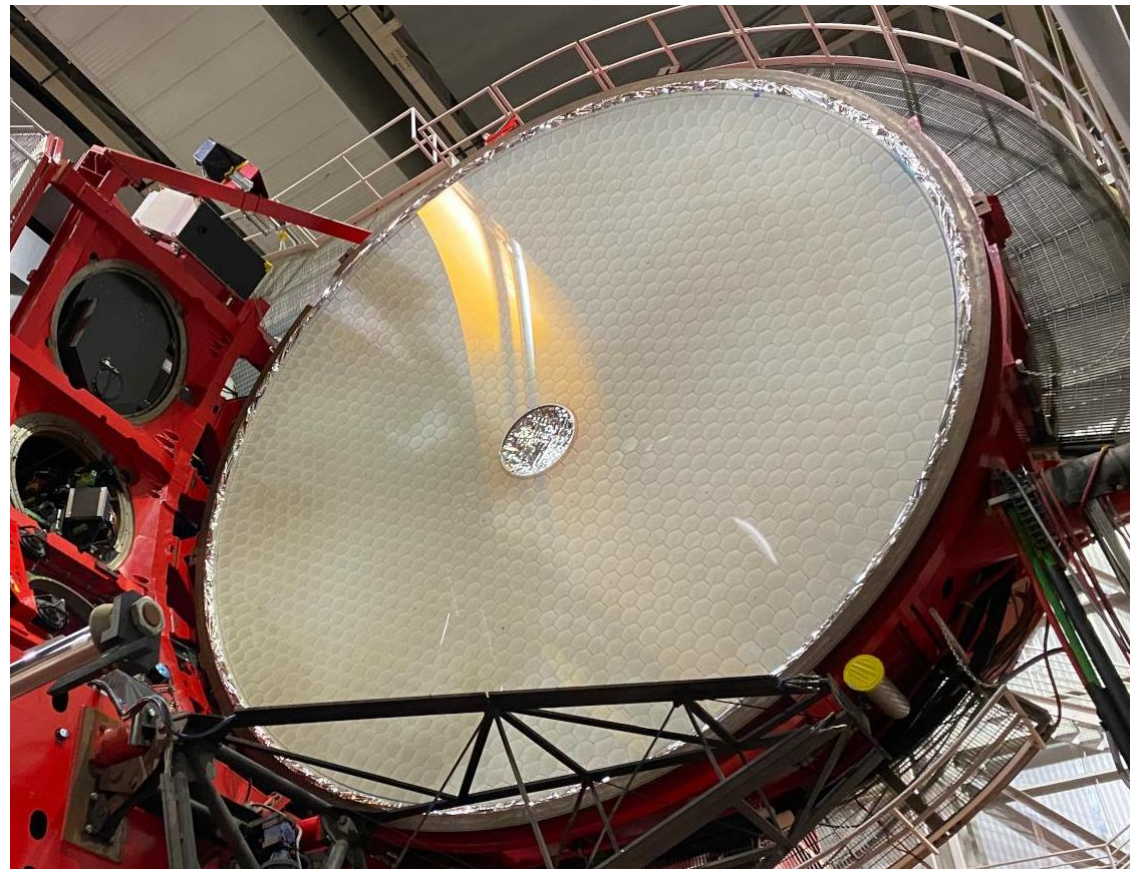
$$m - m_{\text{ref}} = -2.5 \log_{10} \left(\frac{F}{F_{\text{ref}}} \right).$$

Scaling SNR Between Observatories

Keck Telescope (10 meter)



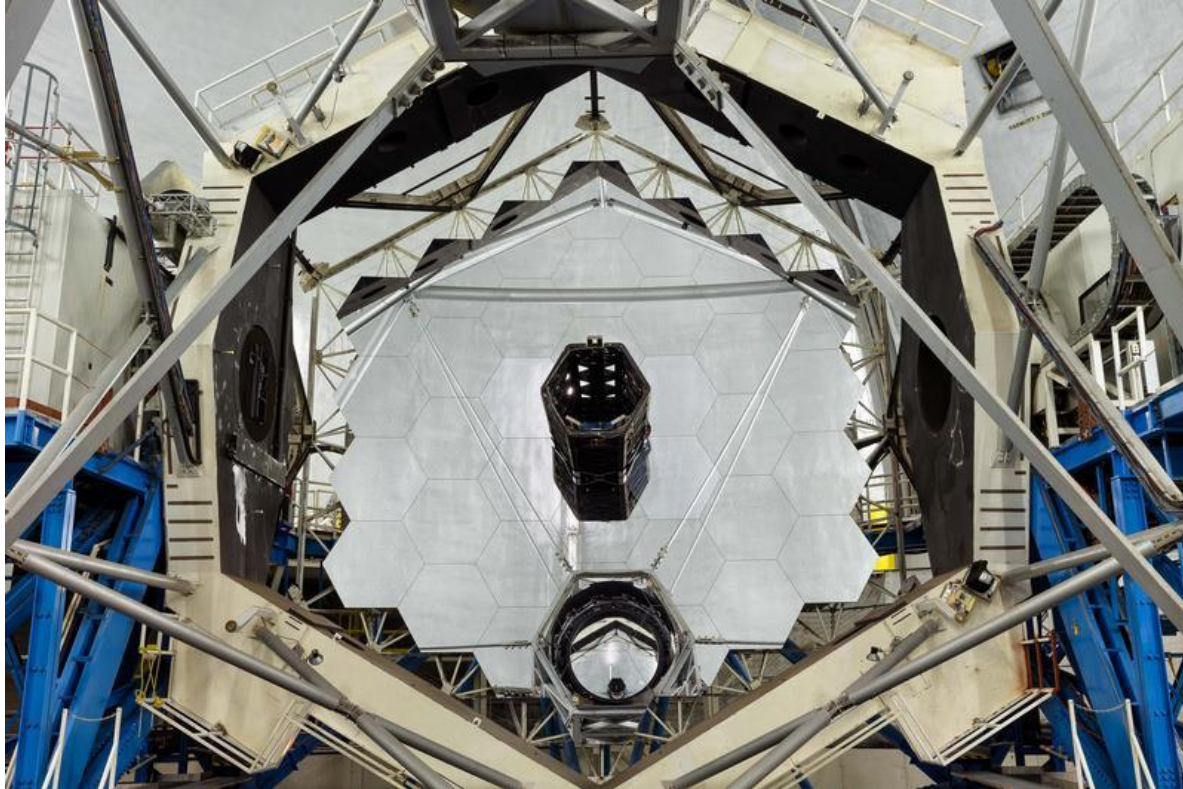
Large Binocular Telescope (8.4 meter)



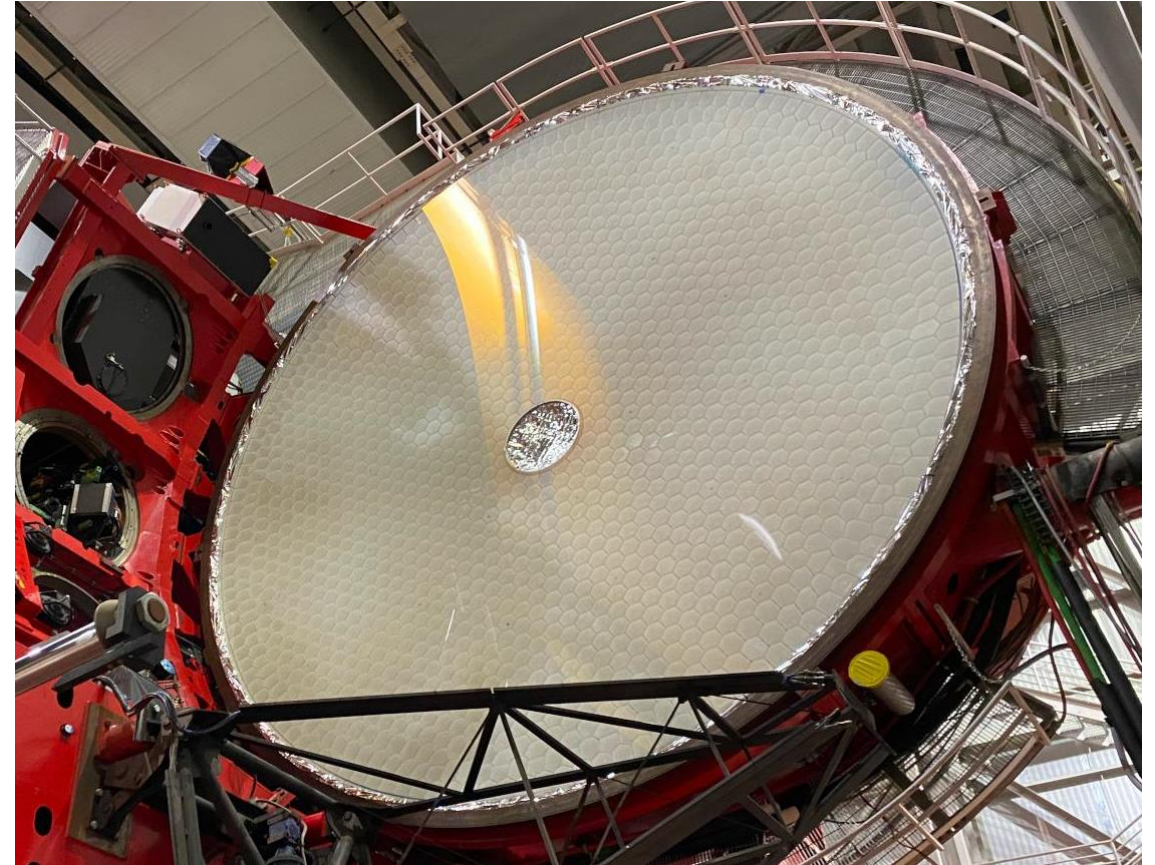
Some instruments do not have an exposure time calculator so you may have to estimate what your observations may look like at a different site/telescope

Scaling SNR Between Observatories

Keck Telescope (10 meter)



Large Binocular Telescope (8.4 meter)



The amount of photons a telescope can detect depends on its collecting area.
The larger a telescope is, the better it is at making precise measurements.

Scaling SNR Between Observatories

$$C_{\text{star telescope2}} = C_{\text{star telescope1}} \times \frac{\text{Area}_{\text{telescope2}}}{\text{Area}_{\text{telescope1}}}$$

Scaling SNR Between Observatories

$$C_{\text{star telescope2}} = C_{\text{star telescope1}} \times \frac{\text{Area}_{\text{telescope2}}}{\text{Area}_{\text{telescope1}}}$$

$$C_{\text{star telescope2}} = C_{\text{star telescope1}} \times \frac{D^2_{\text{telescope2}}}{D^2_{\text{telescope1}}}$$

$$\text{SNR} = \sqrt{C_{\text{star}} \times t}$$

$$\text{SNR} = \sqrt{C_{\text{star}} \frac{D_{\text{telescope2}}^2}{D_{\text{telescope1}}^2} \times t}$$

$$\text{SNR} = \frac{D_{\text{telescope2}}}{D_{\text{telescope1}}} \sqrt{C_{\text{star}} \times t}$$

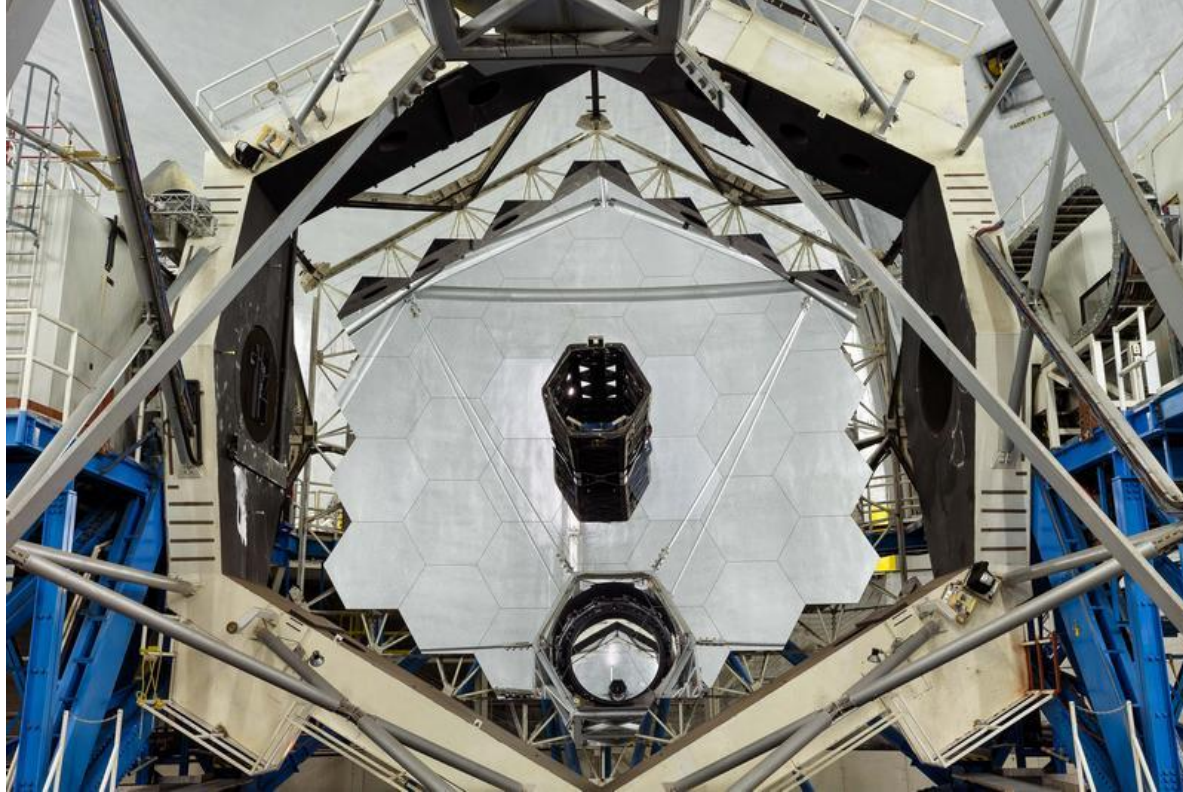
Scaling SNR Between Observatories

$$\text{SNR} = \frac{D_{\text{telescope2}}}{D_{\text{telescope1}}} \sqrt{C_{\text{star}} \times t}$$

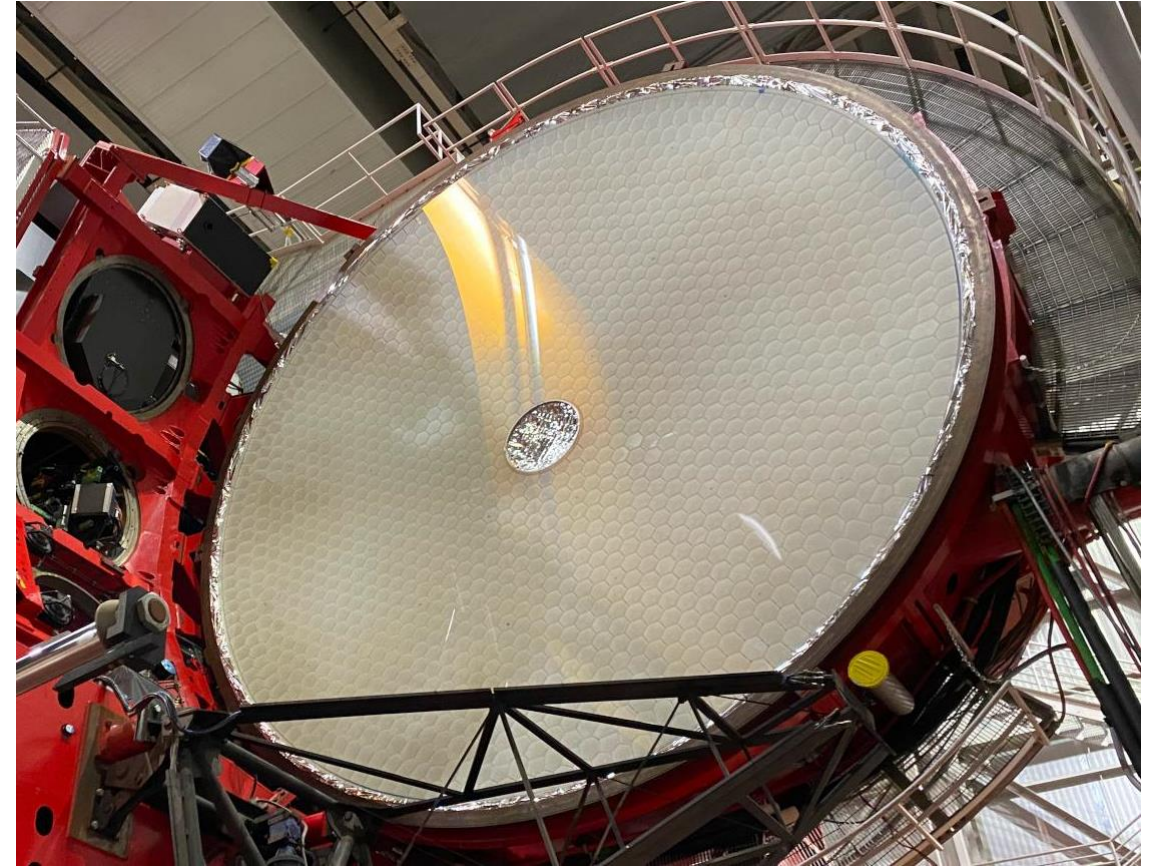
- **IF EVERYTHING ELSE IS CONSTANT**, SNR scales linearly with telescope diameter

Example: Scaling Telescope Diameter

Keck Telescope (10 meter)



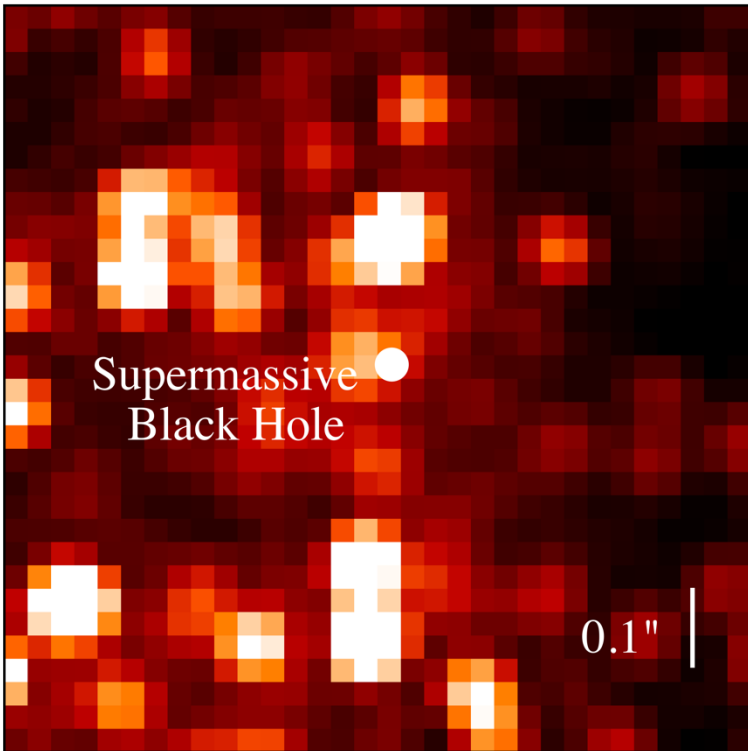
Large Binocular Telescope (8.4 meter)



Caveats: Scaling Telescope Diameter

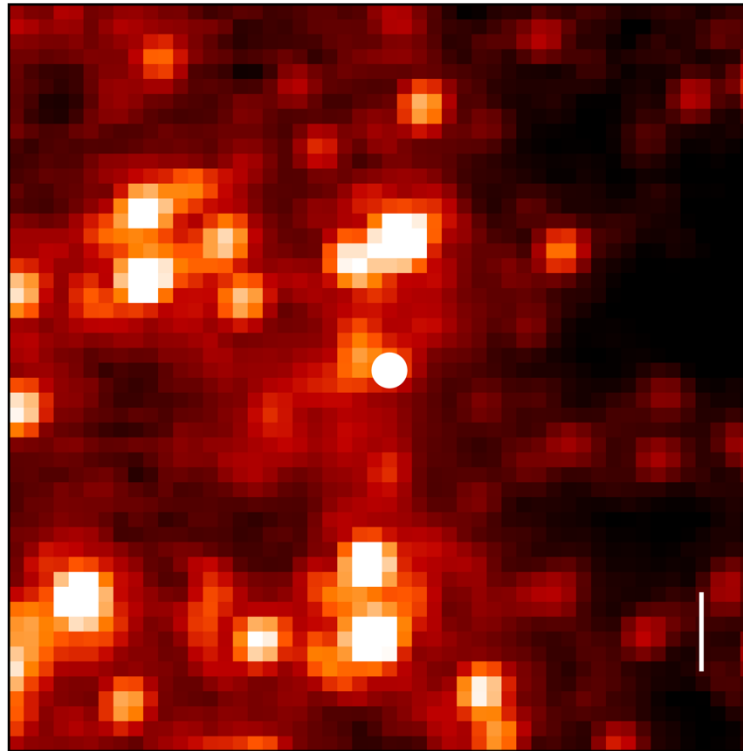
Galactic Center's Central Arcsecond

James Webb Space Telescope



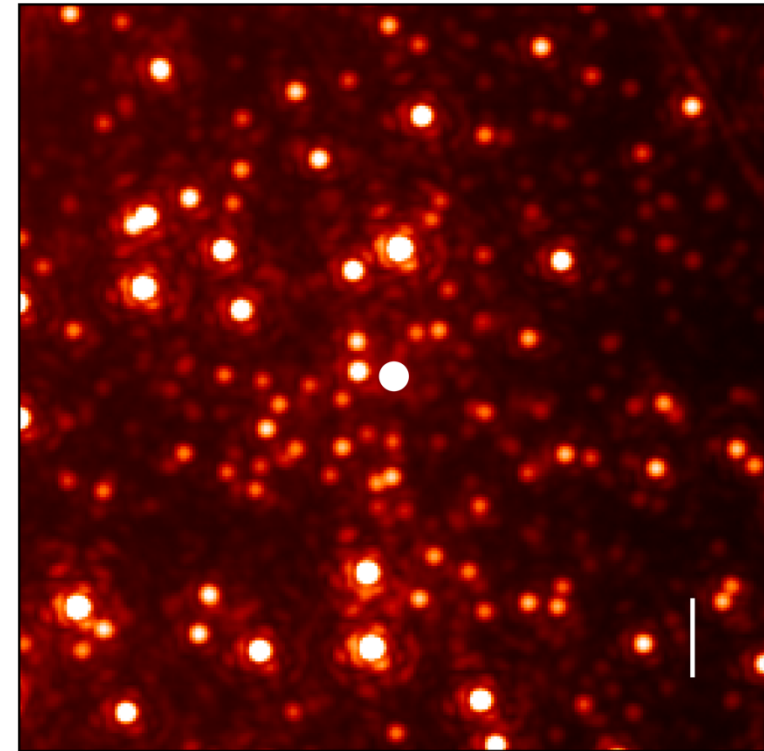
6.5 meter

Current Facility
(Keck Observatory)



10 meter

Thirty Meter Telescope



30 meter

Figure from IRIS Team (PI: Shelley Wright)